

Wednesday Sept2nd

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1 Last class summary

We were talking about linear regression using gradients.

↑ Price → size ↓ Bedroom

Prediction = $\omega_0 + \omega_1 * \text{size} + \omega_2 * \text{BR}$ — $[\omega_0, \omega_1, \omega_2]$ Error = $1/2n$

$\sum_{i=1}^n [\text{Predicted}^i - \text{actual}^i]^2$

Minumum error Algorithm → gradient descent $[\omega_0(\text{hat}), \omega_1(\text{hat}), \omega_2(\text{hat})]$

Evaluation: Train Dataset ✓ Test Dataset we usually keep this to confirm our results

Mean Square Error for test dataset. Hopefully it is low enough so we can accept the model. Mean Absolute Error Mean Absolute Percentage Error

In homework we are allowed to use mean squared error. Here is the thing, it does not mean much by itself, if i get a answer of 100 is it good or bad. Unless you know the scale, you cannot know if it bad or good.

Mean absolute percentage error. Another way to measure the performance of a model, the model can try to trick you if it has seen the data. Keep a cast data,

train: what the model has been trained on validation: dataset under your control that you use to test and evaluate the data. test: in production, what is the error when it is deployed into production. Tested out in the wild

2 Suppose asked to build a model of the daily temperature prediction

January 1st ———> December 31st

There will be a column for actual and predicted data | Actual | Predicted |

Variance, how much the data varies below the mean.

Somewhere in our data we have -2, and 118 this is a pretty big spread. Lets assumed the average is 70. Somebody gives us a prediction model thats pretty dumb but everyday it says 70. They claim my average is the actual average “so it is perfect”. What is the flaw in this argument, the problem is that his model does not account for **variance**

Variance = $\sum(x_i - \bar{x})^2$ What is the real variance, high to low. The variance of this model is 0, it is a very poor job in the data. **fraction of variance in the data that is explained as: R^2 value**

$R^2 = 1 - \sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y})^2$ —> $\hat{y}_i$ predicted value ; is the average value

HOMEWORK

we will choose a dataset

UCIM repository. Then search for regression task. Get familiar with data sources. Kaggle, you will have to preprocess the data, handle null values think of what to do.

Split the data into 2 parts, sklearn it has split and use that. training part and testing part. 80 20 or 70 30 you decide. Build your model using gradient descent going to have to code gradient descent from scratch.

100 page machine language book it is free and you can use it to learn, has a github and if you go there you can see that they have the gradient descent coded for you. It is coded with only one variable, you will have to use multiple. Line 40 update w line 22 w learning rate α pretty simple. Code this algorithm from scratch. Learning rate is the step size and you will have to try various values of the learning grid. Line 60 for learning rate.

Output as many evaluation statistics as possible.

2nd part we can use built in libraries.

SGDRegressor in this part we can use this built in library and then compare the results whether it is bad or good.

Linear regression is an **Idealized** model

- Salary = 60,000 + 20,000 * [yrs_{worked}] do not blame yourself if your model does not perform that well

Everything works in a linear model.

3 Decision Tree Model:

^ humidity $\wedge$ <80% $\geq$ 80%

Tree model is loved by computer scientists. Lets begin with a simple model The tree growth will stop when you have reached only a single class, right now there is a mixture. I will keep splitting the tree until i have a clear data

Tree keeps on splitting until you have pure nodes until single class + or -

once the tree is built, you can travel down the tree until you get to a leaf node until correct

Most of the cases you will only split it binary -> into two parts

4 Learning a DT

What variable should be on top of the other?

Root nodes and leaf nodes, internal nodes. How do we represent the splits? Split whether x2 is below 3/

We will continue splitting until the data is clear, on this side this happens on this side something happens. Keep doing until each region is pure.

4.1 Finding the best split

How do i determine whether a prediciton will be the correct choice?

What is the best factor that tells me my answer.

We need to measure purity and impurity we need to find numerically what is each.

Entropy - this is related to data science. Entropy of a mixture. Suppose we have two classes , 2 or more classes in a data is taken in each class and then find out the probablitiy of that class in this data set

$$P_1 = 16/30 \quad P_2 = 14/30$$

$$\mu = \sum - p_i \log p_i$$

Entropy is a degree of mixing, the more pure the data is, the lower the entropy is.

Entropy Algorithm: try all of the features, the one that reduces entropy the most is the one we will choose. Just like gradient descent, we want to go down the path from high to low. Machine learning is all about optimization down until we reach lowest ideally 0.

Information Gain = $H_p - \text{average}(c_1, c_2) \cdot n_1/n_1+n_2$ inside the average is the entropy. How much reduction in entropy before the last split. Information gain means how much reduction in entropy has happened because of this split. Do we want information gain to be low or high???

We want it to be high because we want a high reduction in entropy.

Algorithm will be as simple as that max(IG) find the attribute that maximizes the information gain, if the entropy is less than 0 again split, keep doing this until we reach 0 entropy. Then we will have a tree that reduces entropy.